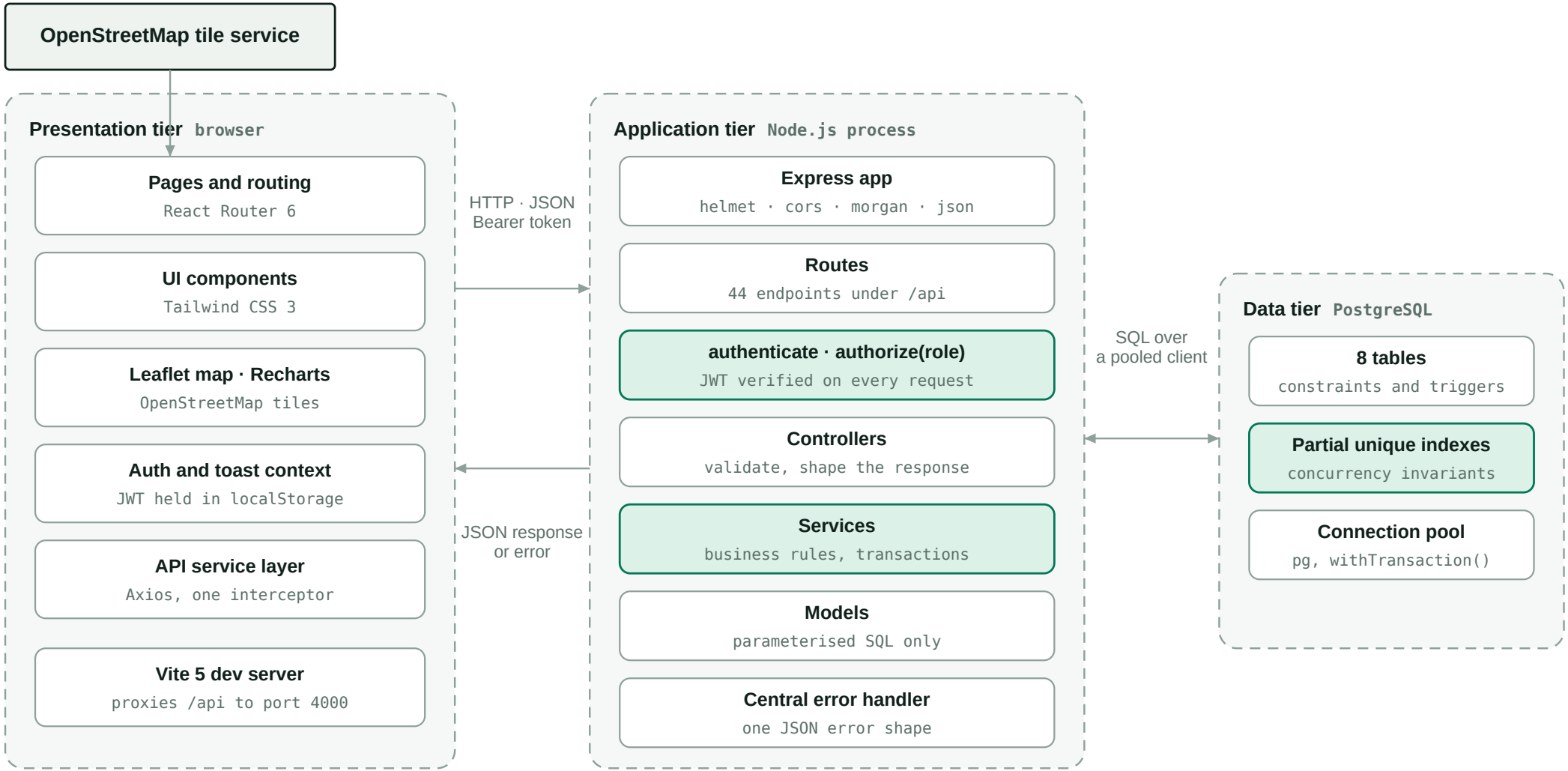
OpenStreetMap tile service
Presentation tier browser
Pages and routing
React Router 6
UI components
Tailwind CSS 3
Leaflet map · Recharts
OpenStreetMap tiles
Auth and toast context
JWT held in localStorage
API service layer
Axios, one interceptor
Vite 5 dev server
proxies /api to port 4000
HTTP · JSON
Bearer token
JSON response
or error
Application tier Node.js process
Express app
helmet · cors · morgan · json
Routes
44 endpoints under /api
authenticate · authorize(role)
JWT verified on every request
Controllers
validate, shape the response
Services
business rules, transactions
Models
parameterised SQL only
Central error handler
one JSON error shape
SQL over
a pooled client
Data tier PostgreSQL
8 tables
constraints and triggers
Partial unique indexes
concurrency invariants
Connection pool
pg, withTransaction()